Manav Kaushal

AI Systems Engineer — LLM agents, evaluation, and backend reliability

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Experience

Core Engineer May 2026 – Present

Analytics Depot — AI analytics platform Remote

- Rebuilt the dataset ingestion pipeline with async parallelism, Redis caching, lazy loading, and content-hash LLM caching, cutting **50K-row upload time from ~124s (timing out) to 39.2s** and database insertion from **19.7s to 1.1s**; fixed a P0 bug that silently dropped file-attached messages.
- Rebuilt chart-accuracy evaluation after finding the existing scorer rated a server that drew *no* charts at **78.1%** (now 0%); froze a 78-case hash-verified acceptance suite and bound charts to the exact computed result, failing closed instead of truncating, re-sorting, or substituting data.
- Built agent observability and evaluation infrastructure: tenant-scoped execution tracing, billed-vs-attributable cost accounting, a provider circuit breaker, and a **blocking cost-regression CI gate**; raised seeded-mutation detection from 7/9 to **9/9 with 0% false failures**.
- Shipped **Saved Metrics**, letting users define a business calculation once and reuse it in agent and NL-to-SQL answers by injecting the saved SQL verbatim as a CTE; delivered 5 rate-limited APIs with zero regressions across 869 existing tests.

Machine Learning Engineer (promoted from ML Intern) Apr 2025 – Nov 2025

Global Village Foundation (GVF) New Delhi, India

- Built a legal-document **RAG system** with Cohere embeddings, FAISS retrieval, and Gemini page-cited summaries, cutting review of a 100-page filing from ~**60 minutes to under 3 minutes**.
- Owned the full application: a FastAPI backend with async PDF ingestion, SQLAlchemy persistence, and Supabase JWT auth, plus a React/TypeScript client with live processing states, serving multi-user document and summary history.

Projects

Replex — Agentic Product-Demo Video Compiler github.com/maybemnv/Replex

TypeScript · Playwright · FFmpeg · Gemini · Zod · Vitest Sep 2026 – Present

- Compiles an approved Playwright flow and a feature brief into a reproducible product-demo video; when the product changes, it **re-captures only the affected scene** instead of re-recording the whole video.
- Constrained the AI editor to **8 typed editing tools** applied through one validated, pure reducer, so the model never writes shell commands, FFmpeg filtergraphs, or project state directly.
- Verifies every render (checkpoints, blank/frozen frames, duration, 1080p/30fps decode) and hash-proves that a recapture leaves unrelated scenes untouched; tested against 3 adversarial browser apps.

GrabPic — Selfie-Based Event Photo Finder github.com/maybemnv/GrabPic

Next.js · Cloudflare Workers · Hono · R2 · Modal (GPU) · FaceNet · Turso

- Lets event attendees find every photo of themselves from a single selfie, returning matches in **under 5 seconds** instead of scrolling a full event gallery.
- Designed a serverless pipeline: Workers issue signed uploads to R2, Modal GPUs generate 512-dimensional FaceNet embeddings, and per-event vector search keeps each event's faces isolated.
- Secured organizer access and protected asset delivery, with automated cleanup of expired event data.

Technical Skills

- **AI Systems & Evaluation:** LangGraph, LangChain, RAG, agent tracing and evaluation, OpenAI / Claude / Gemini APIs, FAISS, PyTorch, Scikit-learn
- **Backend & Data:** Python, SQL, FastAPI, PostgreSQL, SQLAlchemy, Redis, Celery, async I/O, Pandas
- **Infrastructure:** Docker, GitHub Actions, Cloudflare Workers, R2, Modal, Linux
- **Frontend:** TypeScript, React, Next.js

Education

Master of Computer Applications (MCA), Hybrid — IIT Patna 2027 – 2029 (incoming)

In partnership with IIIT Ranchi

Bachelor of Computer Applications (BCA), Data Science 2022 – 2025

Jagannath Institute of Management and Science, Delhi CGPA: 7.41 / 10

Achievements

- Won **3 first-place finishes** across 10+ national-level AI/ML and systems hackathons; invited back to judge and mentor teams at **5+ hackathons**.
- Mentored **100+ students** on Python, Git, system design, and AI-assisted development with Codex and Claude Code.